

RRSS	Version: 1.0
Software Design Description	Date: 2024/05/9

RRSS

Software Design Description

1. Revision History

<i>Version</i>	<i>Date</i>	<i>Author</i>	<i>Change Description</i>
0.1	13.05.2024	Eyüp Menevşe	Introduction and System Overview addition.
0.2	14.05.2024	İlker Avcı	Added updated class diagram
0.3	14.05.2024	İlker Avcı	Added updated ER diagram
0.4	15.05.2024	Buğra Kağan Acar	Added "Manage account" sequence diagram
0.5	16.05.2024	İlker Avcı	Added "Search product" sequence diagram
0.6	17.05.2024	Eyüp Menevşe	Added "Manage product" sequence diagram
0.7	18.05.2024	Mete Uysal	Added "View analytics", "Chat with ChatBot", "Reply to review" sequence diagrams
0.8	18.05.2024	Furkan Şimşekli	Added "Manage review", "Manage topic", "Manage topic entry" sequence diagrams
0.9	19.05.2024	Mete Uysal	Added state chart for "Review"
0.10	19.05.2024	Mete Uysal	Wrote "5. Requirement Traceability" section
0.11	19.05.2024	Furkan Şimşekli	Wrote verbose part of "4. Design Details" section
1.0	19.05.2024	Furkan Şimşekli	Refactored document

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2. Introduction

2.1 Purpose and Scope

This document contains in-depth information about the design of the software project RRSS(Review and Recommendation System for ShopSmart). There are different types of information used such as diagrams and visual designs to support this document. Overall, this document provides our project's design information that would meet the requirements specified in the System-Wide Requirements Specification document.

2.2 Document Overview

This document provides information about these chapters:

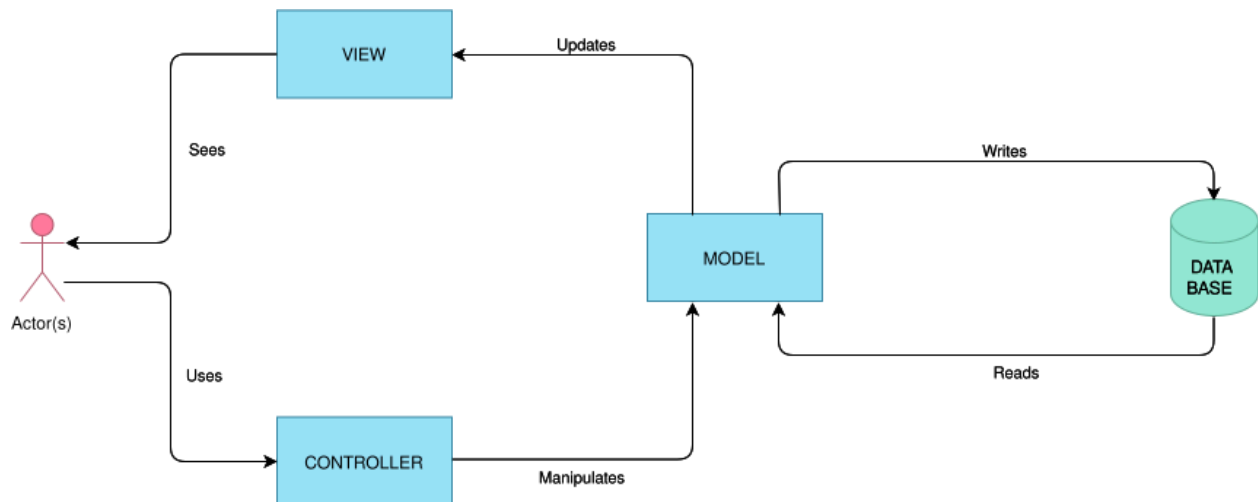
- Revision History - Changes to this document are recorded in this chapter.
- Introduction - General information about the document is given in this chapter. This chapter is divided into 5 sub-chapters. In purpose and scope, a general short definition of this document is given. In the document overview, the chapters and information about them are described. In system overview, the overall system is described and supported with visual(s). In definitions, acronyms and abbreviations; the terminology specific for this document(not standard) are defined. In the references, the references used in the document are given.
- Design Constraints and Decisions - Description of the general constraints of the design process and decisions about the design process are given in this chapter.
- Design Details - The drawings which are Class diagram, Sequence diagrams for use-cases specified on System-Wide Requirements Specification document, Entity-Relationship diagram, and user interfaces are given in this chapter.
- Requirements Traceability - A traceability relationship between the requirements and software design is given in this chapter using a table to ensure all specifications are fulfilled.
- Annexes - In case there are parts in the documents that need to be explained further, they are given in this chapter.

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2.3 System Overview

RRSS is an online rating and recommendation web platform for e-commerce. The app creates a space for the e-commerce users to see and make reviews, rating about products using their corresponding pages and socialize using community forums.

The project RRSS is built on open source technologies like Spring, SQLite etc. The source code of the project is on GitHub but can not be shared publicly as of now.



MVC pattern is used in this project. The system has Model, View and Controller parts separated. Incoming requests sent by the user from the use cases in the system come to the controller at first. Controller handles the related information by passing it to the View or Model components, gets the results from them and responds to the user. We have an architecture that is structured and easier to understand in creating the system this way.

2.4 Definitions, Acronyms, and Abbreviations

<i>Term/Acronym</i>	<i>Definition</i>
RRSS	Review and Recommendation System for ShopSmart
MVC	Model, View and Controller

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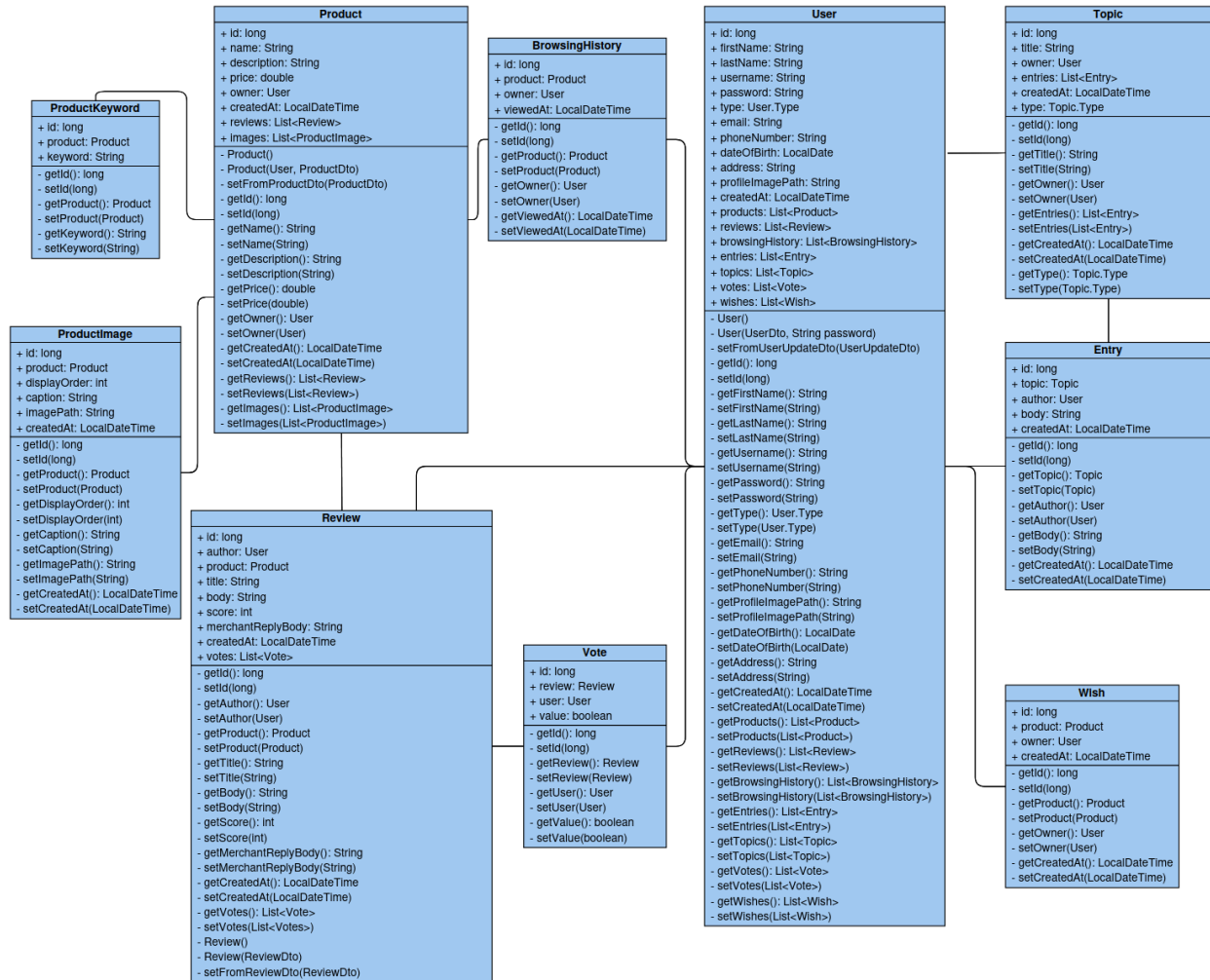
3. Design Constraints and Decisions

- HTML and CSS are used for the frontend and close to no JavaScript or additional libraries are used for the frontend part. This means that the application will be faster as there is close-to no complex computation is made in the frontend part.
- We're using server-side rendering using Thymeleaf to generate HTML pages dynamically from the backend. The user requests and CRUD operations are made using Spring Boot. The user initiates a request from the frontend and the controller is signaled, then the related operation is made on the backend.
- SQLite is used for our database as it's a relational database system and also lightweight. So we can create tables and create relationships between them with each other. Also SQLite does not require a standalone server so it is chosen for this project considering its needs and size.
- MVC pattern is used in this project. The Model, View and Controller components are separated.
- The application is designed to work in most of the browsers including Chromium-based or Firefox, as well as devices/browsers where there is no JavaScript support so overall more than %98 of the browsers are aimed to be supported.
- The system is expected to provide 99% uptime for availability. The application should not be interrupted any longer.
- Obeying the Doherty Threshold, the system is expected to respond in 400 milliseconds at most. The system is developed in a way to obey this so the experience for the user is smooth and uninterrupted.
- To use the system, internet connection is required, as the application is a web application.

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4. Design Details

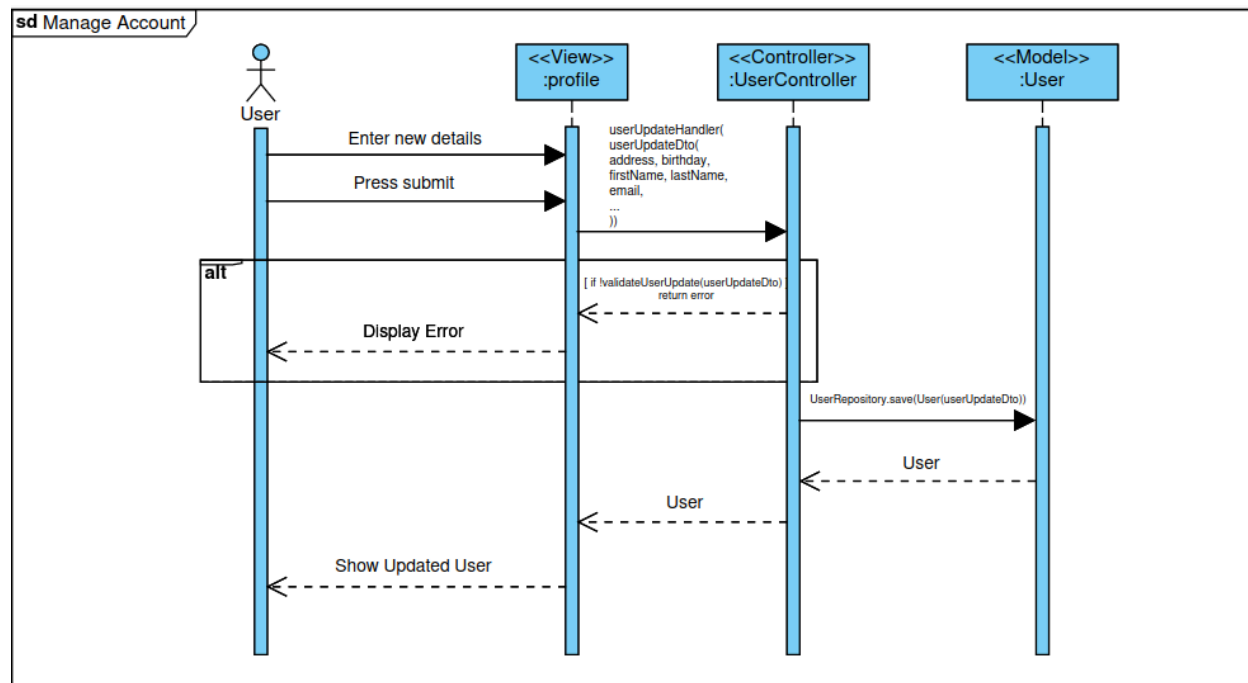
4.1 Software Components



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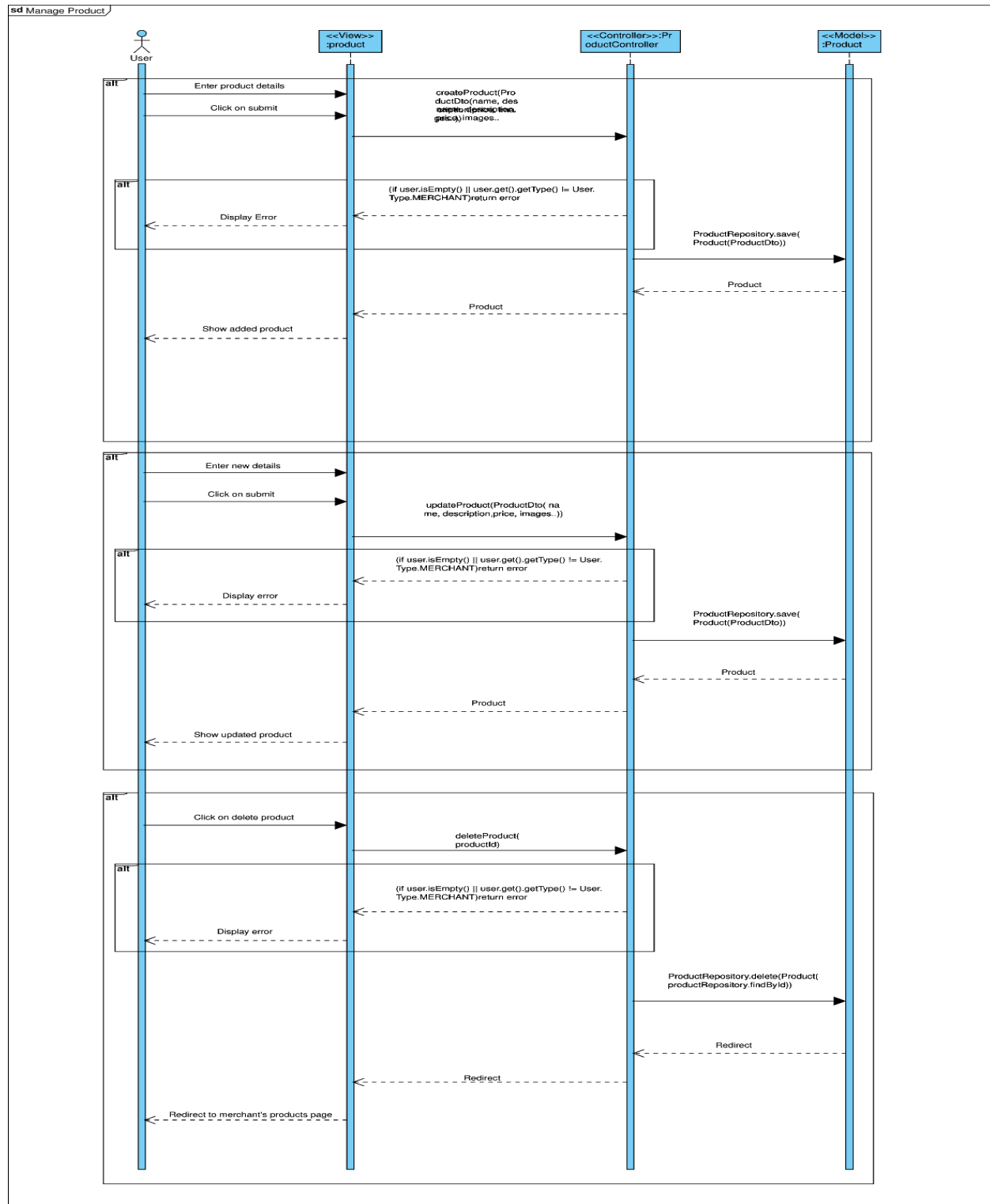
4.2 Software Behavior

“Manage Account” Sequence Diagram:



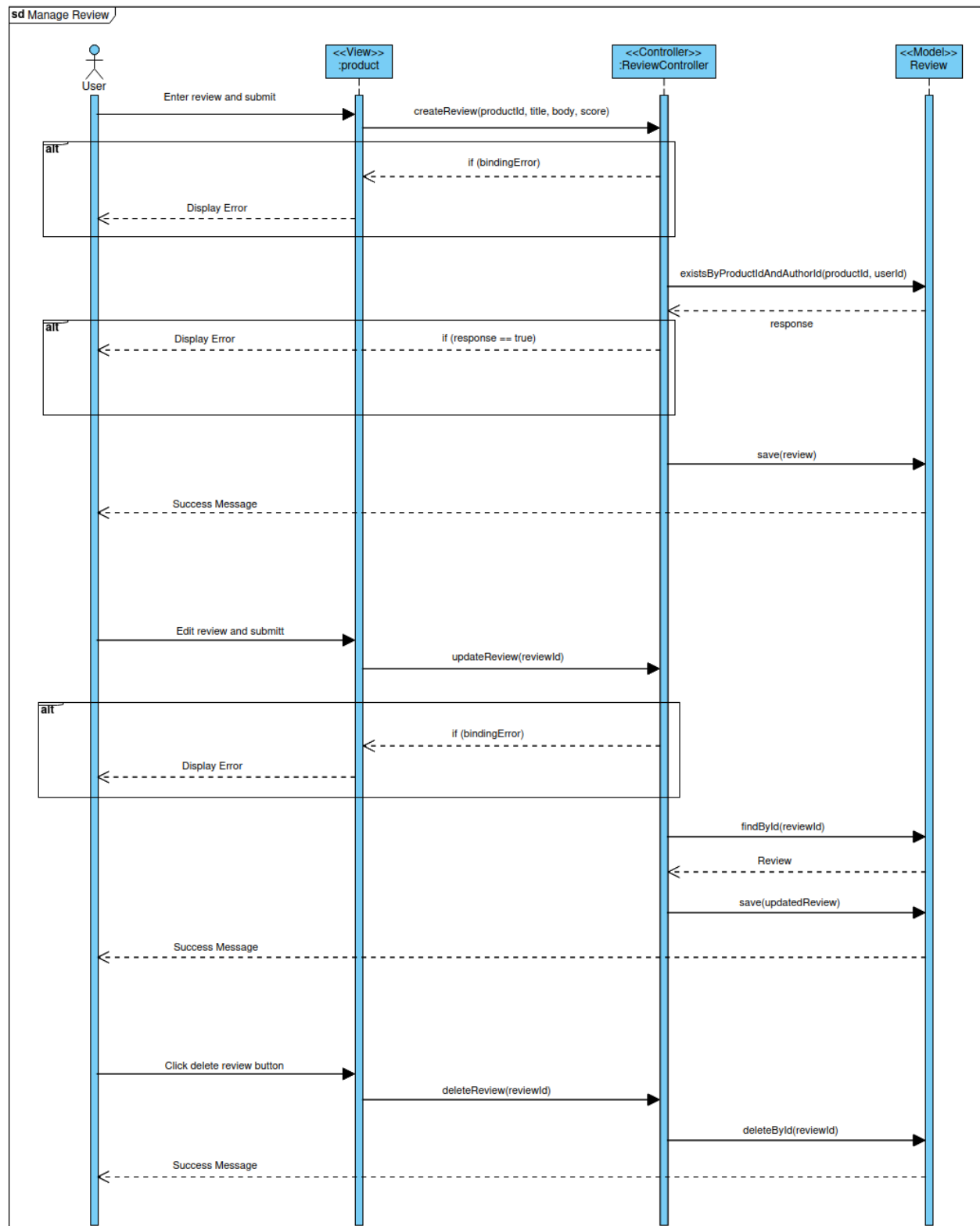
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“Manage Product” Sequence Diagram:



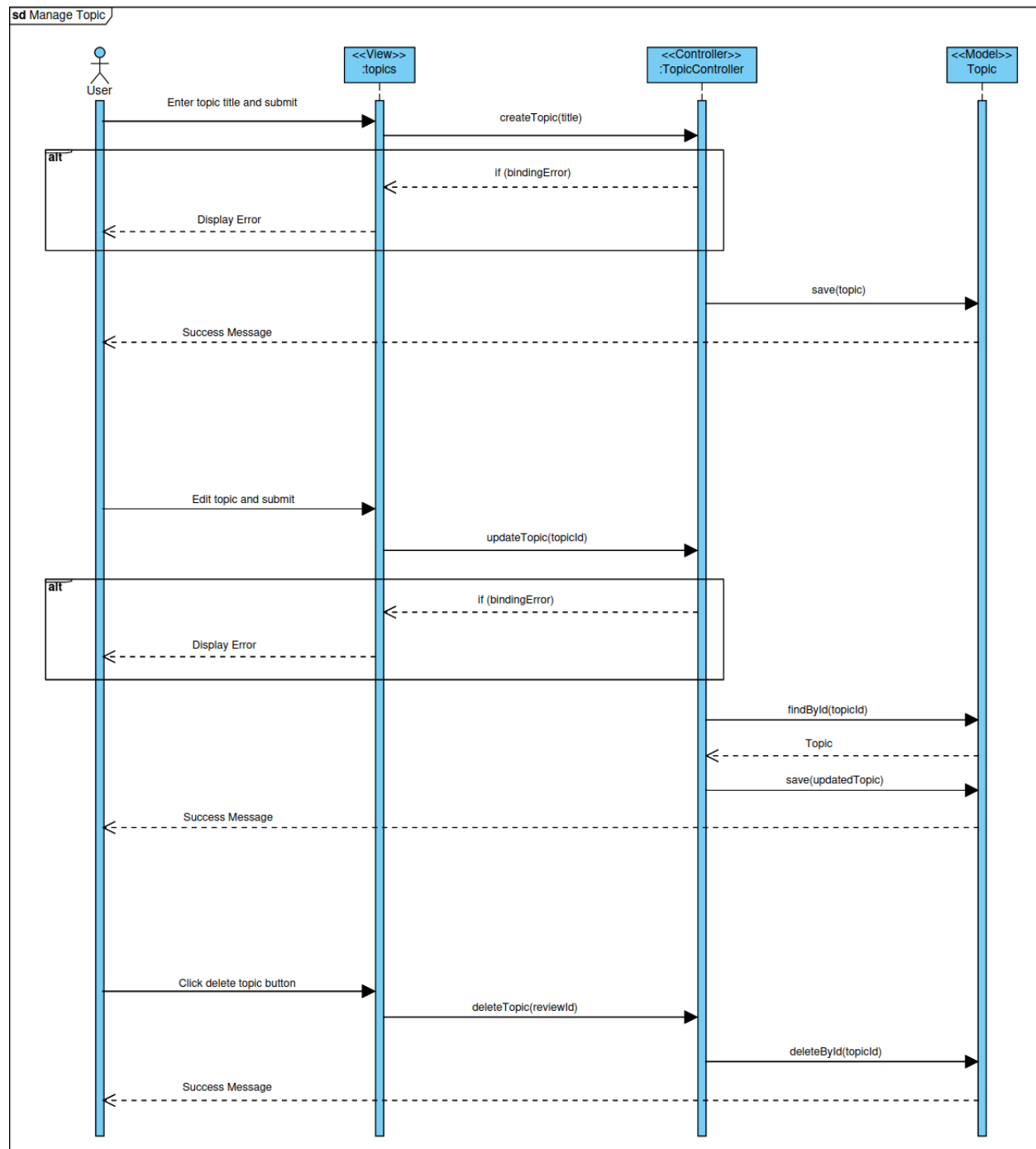
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“Manage Review” Sequence Diagram:



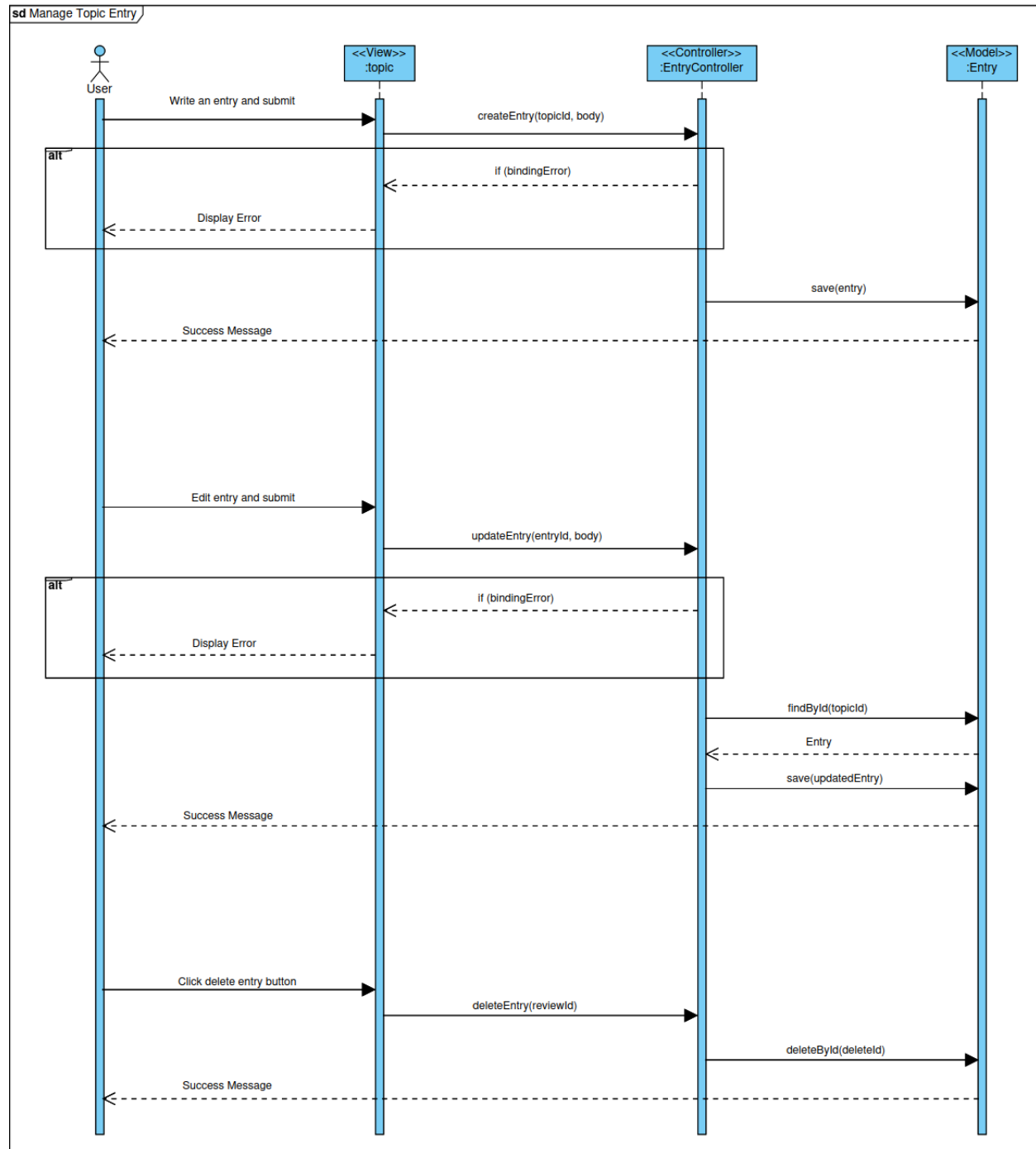
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“Manage Topic” Sequence Diagram:



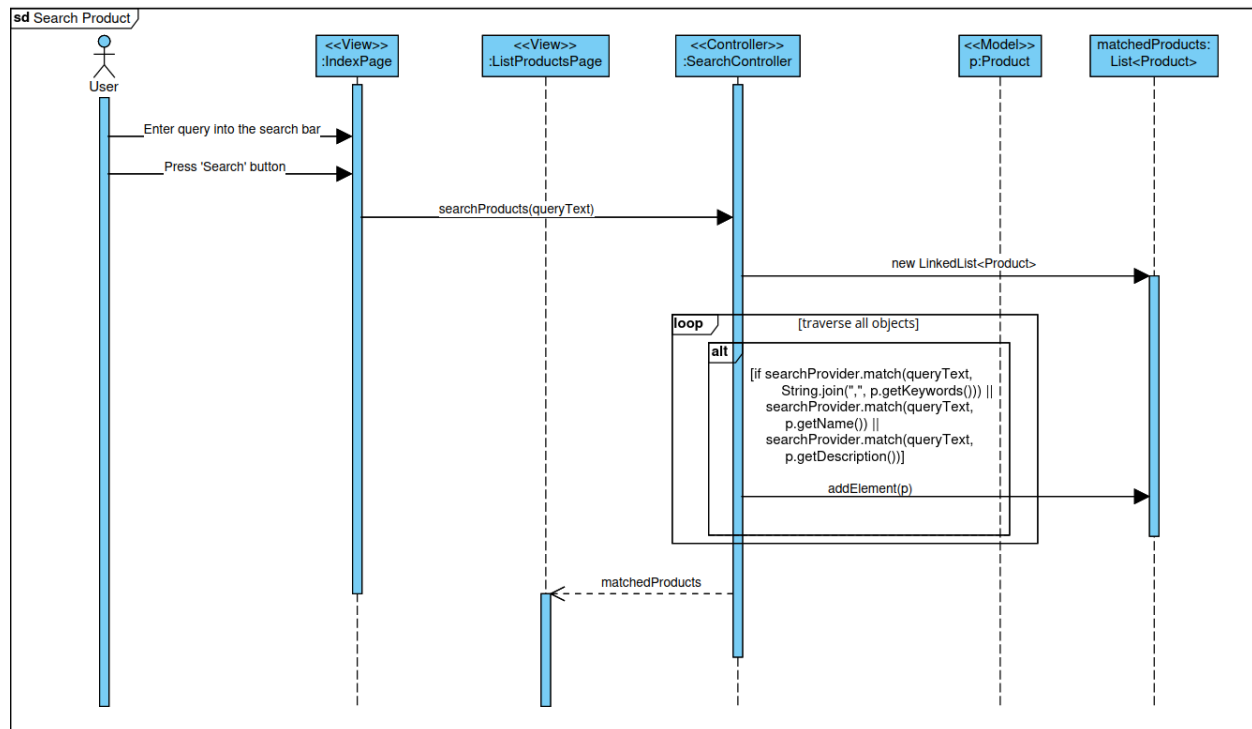
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“Manage Topic Entry” Sequence Diagram:

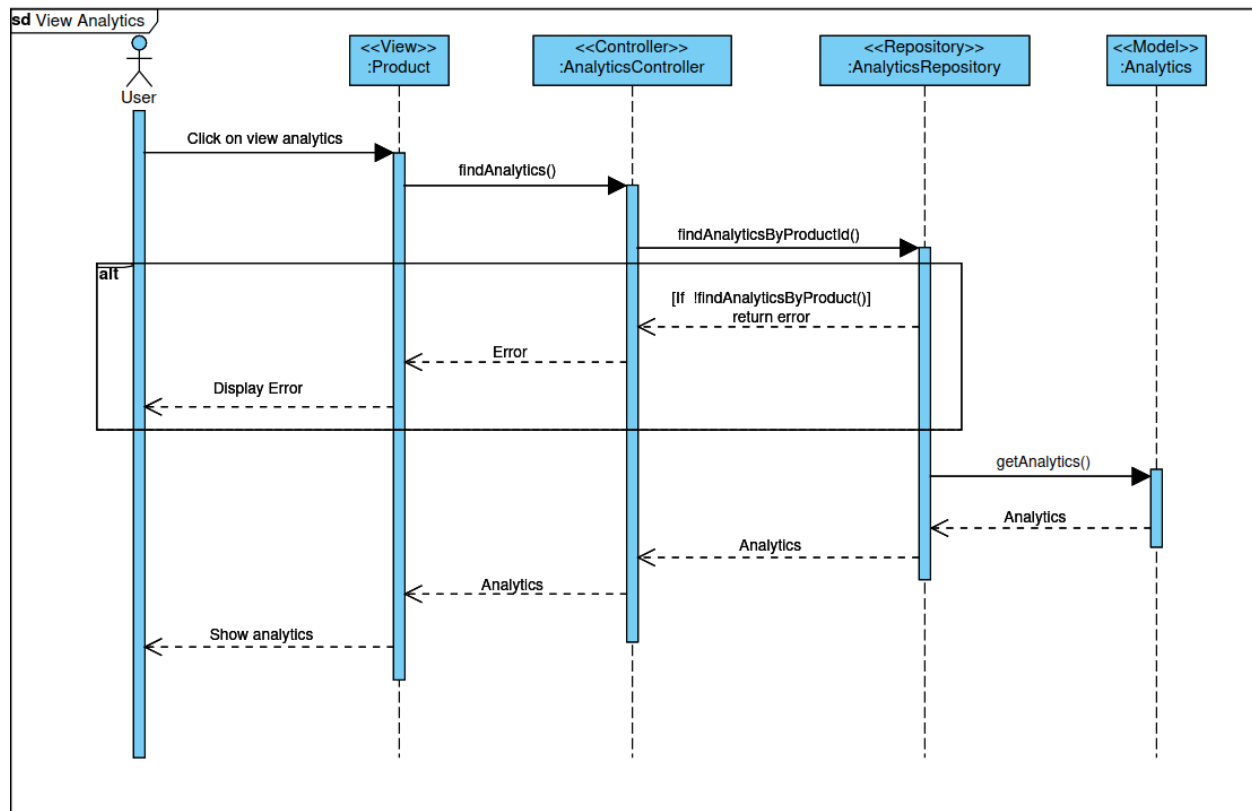


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“Search Product” Sequence Diagram:

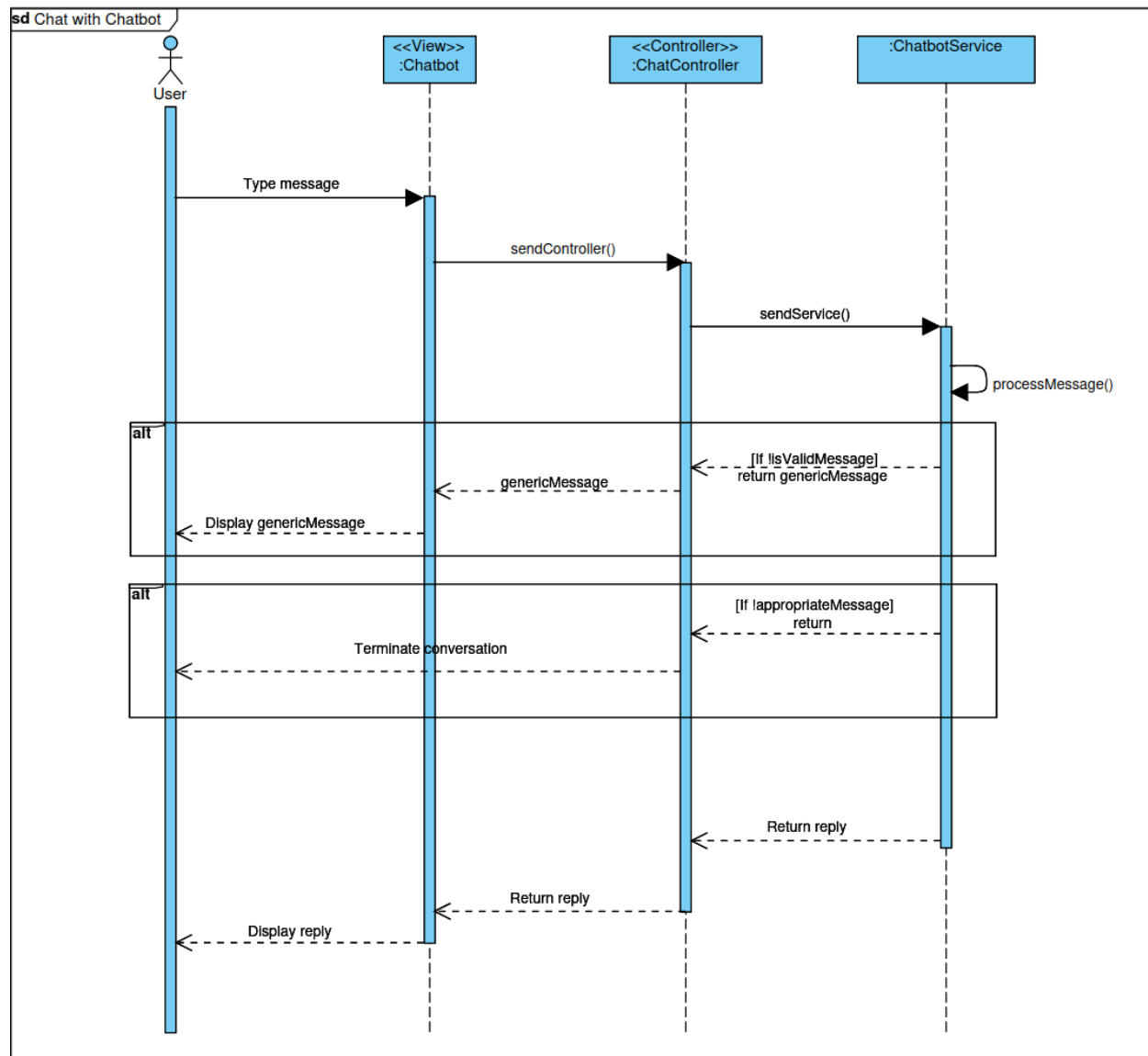


“View Analytics” Sequence Diagram:

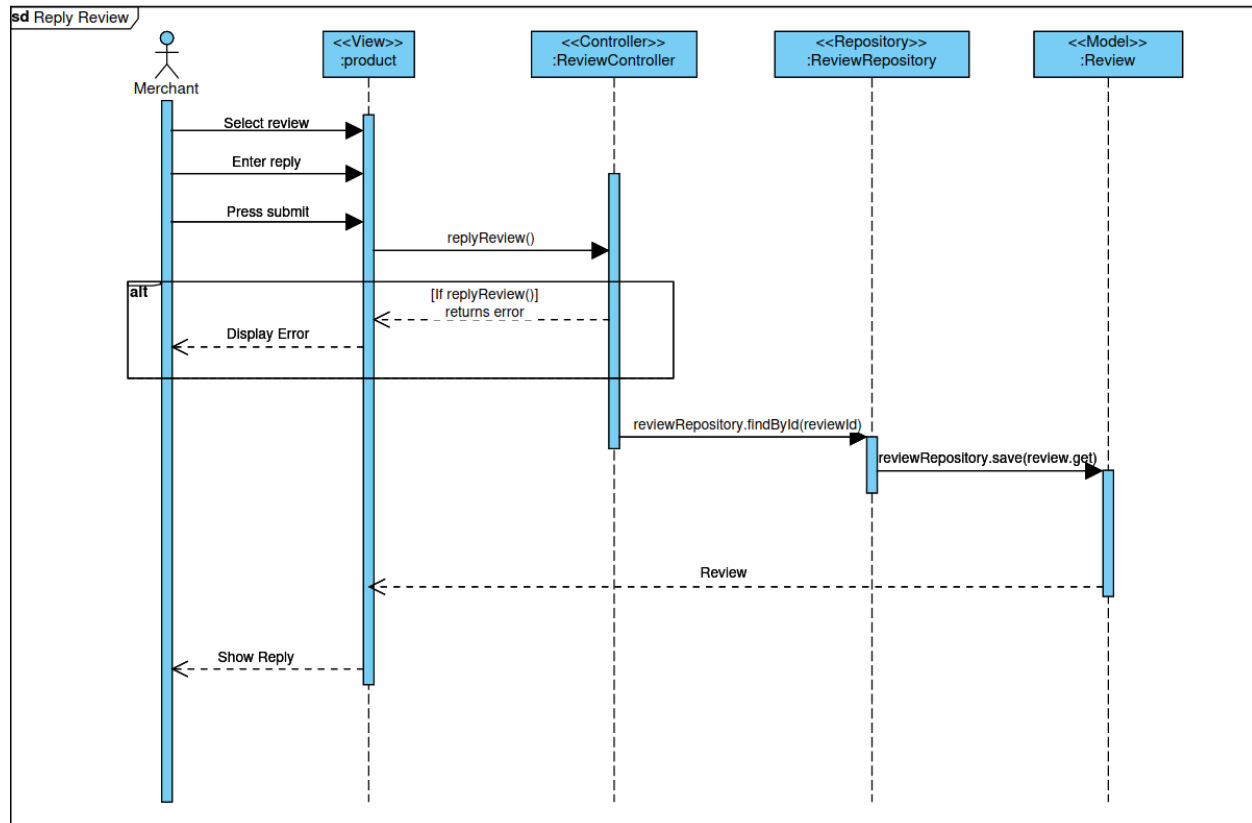


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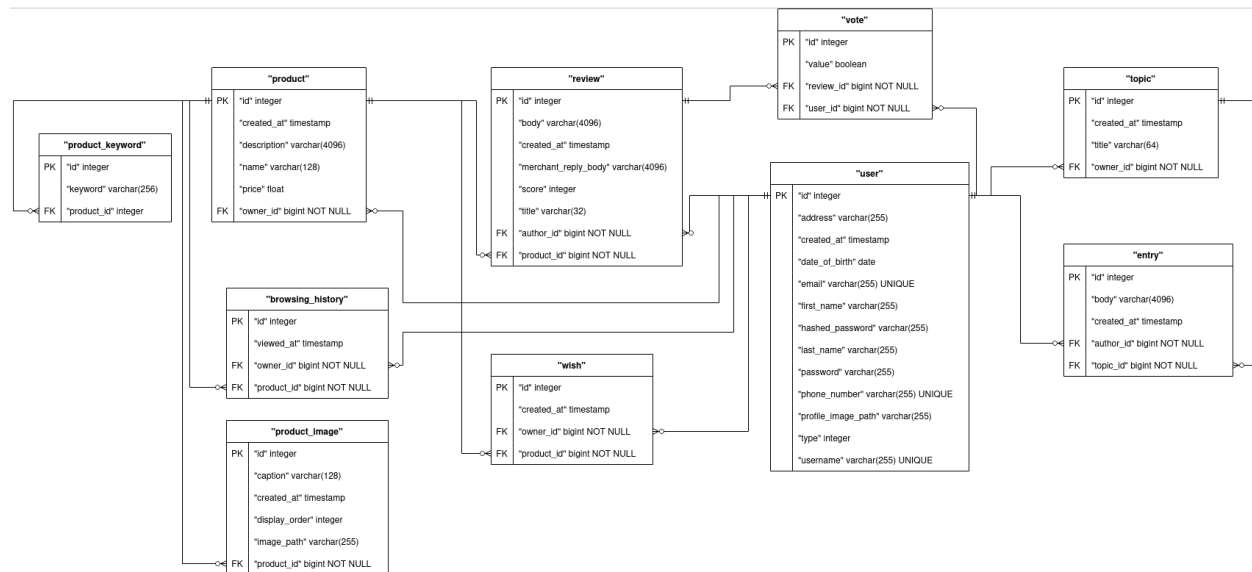
“Chat with ChatBot” Sequence Diagram:



“Reply Review” Sequence Diagram:



4.3 Data Model (E-R Diagram)



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4.4 User Interface Design

Sign Up Page with Errors:



Sign Up

→ Log In

Harold

Last name

Last name must be between 1-64 characters

f1nch

finch@thornhill.com

+19370170375

Password

Passwords do not match!

Enter password again

Customer

Merchant

Sign Up

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Sign Up Approvals from Admins' Point of View:

ShopSmart

Search

Community

admin

These Users Await Your Approval



Example Merchant

@example_merchant

Signed up at 19/05/2024

Approve

Delete



Example Customer

@example_customer

Signed up at 19/05/2024

Approve

Delete



Xavi Hernandez

@xavi

Signed up at 19/05/2024

Approve

Delete



Best Merchant

@best_merchant

Signed up at 19/05/2024

Approve

Delete



Andres Iniesta

@iniesta

Signed up at 19/05/2024

Approve

Delete



Harold Finch

@f1nch

Signed up at 19/05/2024

Approve

Delete

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User Profile Page, while updating:

ShopSmart

Community

finch

Account



No file selected.

Cancel

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Example Product Page, while writing a review:

ShopSmart

Community

finch



iPhone 12 Pro 128 GB Black

example_merchant

789.99TL

New generation of iPhone is at Example Merchant! You can't find a better deal than this!

★★★★☆

3 reviews

Add to wishlist

Reviews

Body

Review length must be between 10-4096 characters

★★★★★



★★★★☆ Is this merchant for real?

I feel like I'm scammed because the phone I got seemed used before, there were tiny little scratches on the screen and the back of the device was full of fingerprints.

example_customer

Like (5)

Share



★★★★☆ Pretty good price for an iPhone

I bought it for my son, and it was pretty nice. Shipment was quick and safe. I think it would have been better if the merchant put more effort for the description.

sourcez

Like (5)

Share



★★★★☆ Storage issue

Am I scammed? I got this model because it had 128 GB of storage but mine has only 107 GB, who would know how many more pictures I could take with missing part...

titata

Like (5)

Share

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List of Products that belongs to a Merchant, merchant's point of view:

 ShopSmart

Community

example_merchant

Your Products

+ New Product

**iPhone 12 Pro 128 GB Black**
789.99\$
Published at 19/05/2024

EditDelete

**Product O**
7.99\$
Published at 19/05/2024

EditDelete

**Matebook 14**
999.98\$
Published at 19/05/2024

EditDelete

**Product Q**
0.03\$
Published at 19/05/2024

EditDelete

**Product G**
0.0\$
Published at 19/05/2024

EditDelete

**Product S**
0.0\$
Published at 19/05/2024

EditDelete

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While updating the details of a product:

 ShopSmart

Community

example_merchant

Edit Your Product

Name

iPhone 12 Pro 128 GB Black

Description

Now generation of iPhone is at Example Merchant! You can't find a better deal than this!!

Price

789.99

Images

Browse...

No files selected.

Submit

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5. Requirements Traceability

Use cases/Classes	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Manage Topic	X	X																		X		X		X		
Manage Topic Entry	X	X																		X		X		X		
Manage Review	X	X				X	X			X									X							
Login	X	X		X	X			X	X			X	X									X				
Manage Account	X	X	X	X	X			X	X			X	X								X					
Manage Product	X	X				X				X	X	X			X		X	X	X	X			X	X		
View Analytics	X	X			X	X	X			X	X	X					X	X	X	X	X		X	X		
Share Review on Social Media	X	X					X			X	X								X			X				
Manage Wishlist	X	X				X			X						X	X	X			X	X	X	X			
Register	X	X		X				X														X				
Search Product	X	X				X			X								X	X				X				
Chat with Chatbot	X	X	X	X	X			X																		X
React to Review	X	X						X		X	X	X					X	X	X	X			X			
Send Password Reset Request	X	X	X	X	X			X					X	X									X			
Reply to review	X	X						X		X	X	X					X	X	X	X			X			

1. RrssApplication
2. DbConfig
3. AdminController
4. AuthenticationController
5. MerchantController
6. ProductController
7. ReviewController
8. UserController
9. ChangePasswordDto
10. ProductDto
11. ReviewDto
12. ReviewReplyDto
13. UserDto
14. UserUpdateDto
15. BrowsingHistory

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16. Entry
17. Product
18. ProductImage
19. ProductKeyword
20. Review
21. Topic
22. User
23. Vote
24. Wish
25. CommunityController
26. ChatContoller

6. Annexes

State diagram for a “Review” can be seen below:

